Scoliosis

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Microbiome Signatures, Microbiome Medicine

Abstract. Scoliosis, a spinal curvature over 10°, is influenced by gut microbiome variations, affecting spinal, respiratory, and cardiovascular health.

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Overview

Scoliosis is a medical condition characterized by an abnormal lateral curvature of the spine, identified when the curvature exceeds a Cobb angle of 10 degrees. It includes types such as idiopathic, congenital, and neuromuscular scoliosis. Idiopathic scoliosis, the most common type, accounts for approximately 80% of cases and mainly affects adolescents, historically with no identifiable cause. Severe cases can lead to complications like spinal cord compression, respiratory difficulties, and cardiovascular issues. Traditional treatments focus on symptom management rather than addressing the root cause. However, recent research has discovered that gut microbiome disturbances are a causal factor in scoliosis development. [1] This insight opens a new frontier for treatment: Microbiome-Targeted Interventions (MBTIs). By correcting gut microbiome imbalances, MBTIs aim to address the root causes of scoliosis, offering a revolutionary approach that could improve patient outcomes and transform scoliosis care.

Causes

Scoliosis, traditionally understood as a musculoskeletal disorder with idiopathic origins, has been found to have a direct causal link with specific gut bacteria, influencing the risk of developing the condition. Recent studies using Mendelian randomization have identified that certain gut microbiota, such as Bilophila and Prevotella9, provide protective effects against scoliosis. Conversely, other bacterial taxa like Mollicutes RF9 and Lachnospiraceae FCS020 group increase the risk of the disease. This groundbreaking discovery suggests that microbial imbalances in the gut could play a significant role in the development of scoliosis, shifting the paradigm from solely structural or genetic causes to include the microbiome's impact. [2]

Interventions

Supplementation should be considered as part of a comprehensive management plan for scoliosis, which may include physical therapy, bracing, microbiome-based interventions, and other treatments as necessary.

▶ **What supplements have support for Scoliosis?**

Research Feed

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October 27, 2023
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Scoliosis major
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Idiopathic
"Idiopathic" is a term used in medicine to describe a disease or condition that arises spontaneously or for which the cause is unknown.

Microbiome-Targeted Interventions (MBTIs)
Microbiome Targeted Interventions (MBTIs) are cutting-edge treatments that utilize information from Microbiome Signatures to modulate the microbiome, revolutionizing medicine with unparalleled precision and impact.

References

1. Causal effects of gut microbiota on scoliosis: A bidirectional two-sample mendelian randomization study.. Lai B, Jiang H, Gao Y, Zhou X.. (Heliyon. (Oct 27. 2023))

2. Causal effects of gut microbiota on scoliosis: A bidirectional two-sample mendelian randomization study.. Lai B, Jiang H, Gao Y, Zhou X.. (Heliyon. (Oct 27. 2023))
3. https://doi.org/.

Lai B, Jiang H, Gao Y, Zhou X.

Causal effects of gut microbiota on scoliosis: A bidirectional two-sample mendelian randomization study. Heliyon. (Oct 27. 2023)

Lai B, Jiang H, Gao Y, Zhou X.

Causal effects of gut microbiota on scoliosis: A bidirectional two-sample mendelian randomization study. Heliyon. (Oct 27. 2023)

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References

1. Causal effects of gut microbiota on scoliosis: A bidirectional two-sample mendelian randomization study.. Lai B, Jiang H, Gao Y, Zhou X.. (Heliyon. (Oct 27. 2023))
2. Causal effects of gut microbiota on scoliosis: A bidirectional two-sample mendelian randomization study.. Lai B, Jiang H, Gao Y, Zhou X.. (Heliyon. (Oct 27. 2023))
3. https://doi.org/.

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